edge-cic

72-pin cart edge, CIClone

File: edge-cic.kicad_sch

rp2354b

RP2354B bus controller

File: rp2354b.kicad_sch

sram

PRG and CHR SRAM

File: sram.kicad_sch

glue

5V glue logic, '595 control bits

File: glue.kicad_sch

esp32s3-sd

ESP32-S3, microSD, status LED

File: esp32s3-sd.kicad_sch

usb-uart

USB-C, CP2102N bridge

File: usb-uart.kicad_sch

power

Power: 5V OR, 3.3V buck, RP LDO

File: power.kicad_sch

FujiNet for the Nintendo Entertainment System - Rev0 (RP2354B)

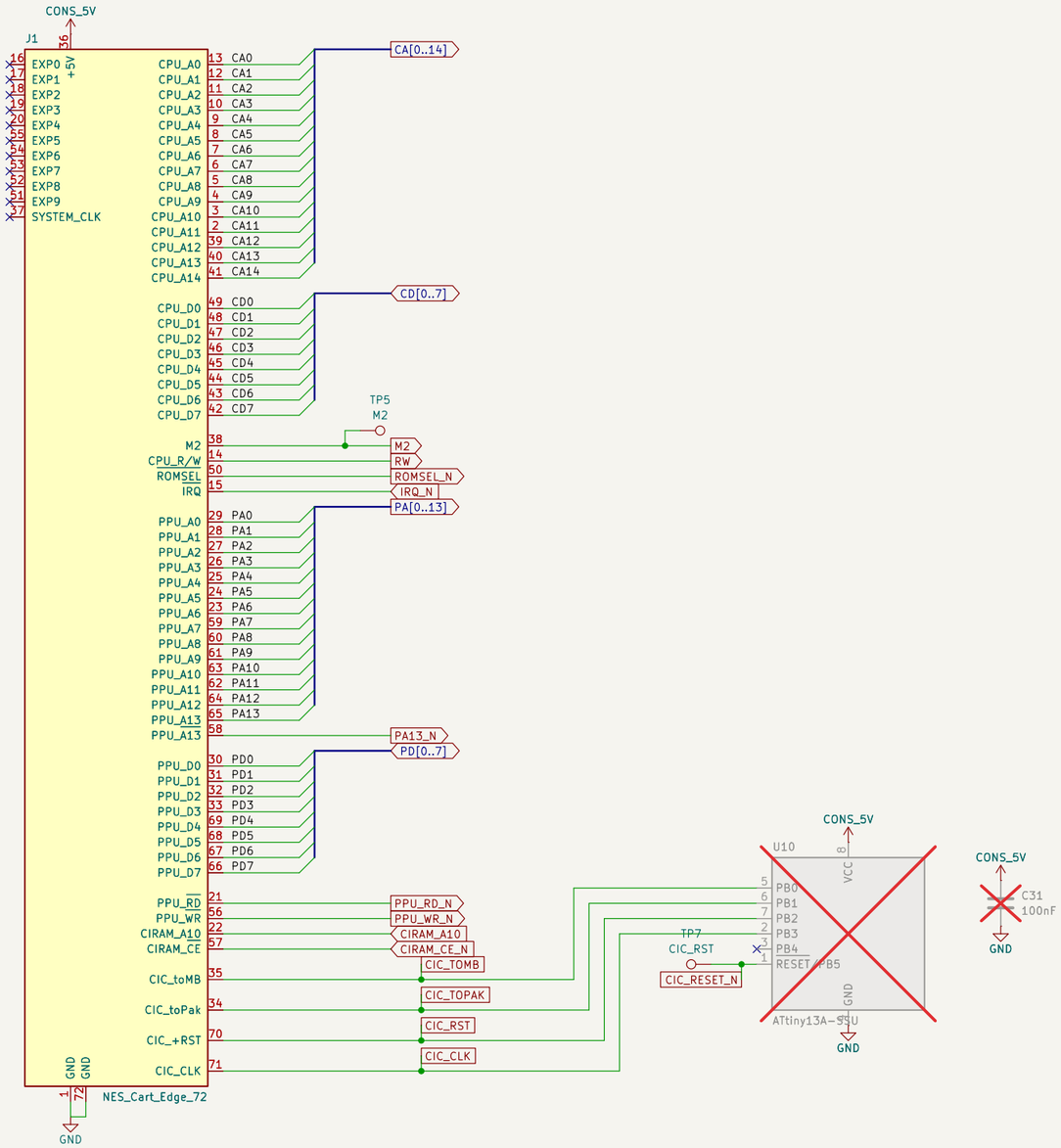
RP2354B (48 GPIO) sits on the 5V cart bus directly through its 5V-tolerant pads: CPU A0-A14, D0-D7, M2, R/W, /ROMSEL, PPU A10-A12 in; /IRQ out; it serves $5000-$7FFF and the reset vectors itself and drives the two AS6C4008 SRAMs' bank lines (PRG A13-A18 = GP33-38, CHR A10-A18 = GP39-47) from PIO lookup tables.
Its IOVDD group runs from a fast LDO (+3V3_RP) so the pads are powered before the console bus is; the core regulator input stays on the buck (+3V3). The SRAMs and the 74HCT glue run on +5V: console 5V through a P-FET (on without USB) ORed with USB VBUS through a Schottky, so their outputs drive the 5V bus at full level.
74HCT595 holds the slow control bits (SRAM_EN PRG_WE_EN CHR_WE_EN MIR0 MIR1 FOURSCREEN LED spare); its /OE is held off for ~140 ms after power-on and whenever the console is off, with 100k pull-downs on every bit.
74HCT253 muxes CIRAM A10 for mirroring and gates CIRAM /CE; PWR_OK (console +5V sense) tri-states both and disables every SRAM output toward the console. PRG /WE = NAND4(M2, ROMSEL, !R/W, PRG_WE_EN) on a 74HCT20: one gate after M2.
ESP32-S3-WROOM-1-N16R8 runs FujiNet (fujiversal-nes) and is the USB host of the RP (CDC, VID 0xCafe);
S3 IO4/IO5 force RP RUN/QSPI_SS so the S3 can re-flash the RP over PICOBOOT.
USB-C: power + CP2102N UART bridge for flashing the S3. microSD on S3 SPI. CIClone (ATtiny13A) on the CIC fingers, DNP until its firmware exists.
Edge: 72 pins, 2.50 mm pitch, 1-36 on the label side, pin 1 at the RIGHT seen from the label side with the fingers down (nesdev Cartridge connector; NES-EWROM-01 geometry). EXP and SYSTEM CLK unconnected.

Generated by tools/gen_sch.py
FujiNet
Sheet: /
File: FujiNet-NES-Rev0.kicad_sch

Title: FujiNet NES Rev0

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 1/8

NES 72-pin edge drawn by function (finger numbers on the pins). Pin 1 is at the RIGHT seen
from the label side with the fingers down (nesdev: NES-EWROM-01 geometry).
EXP0-9 and SYSTEM CLK are not used. The CIClone (ATtiny13A) is DNP until its firmware exists.

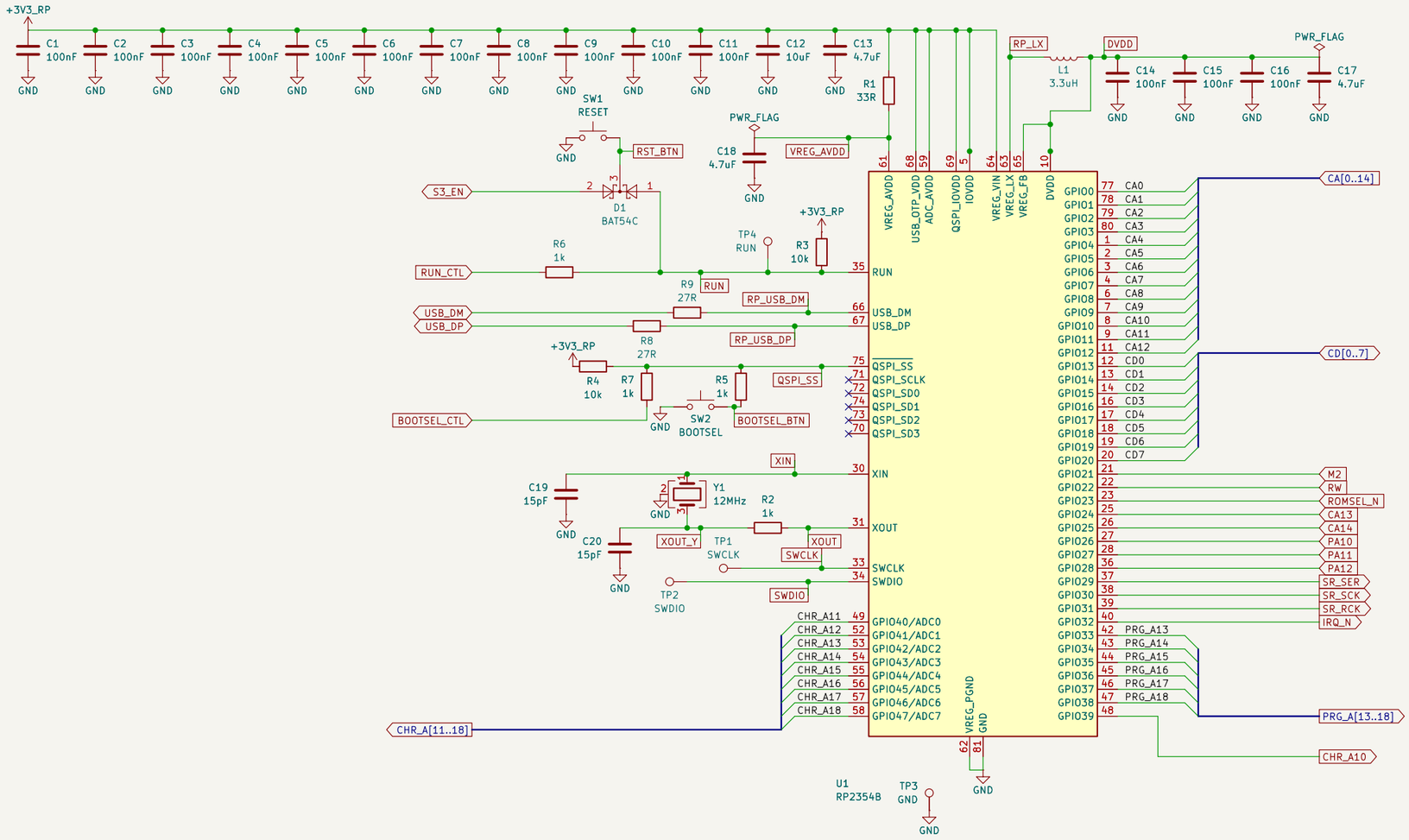


Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file
FujiNet

Sheet: /edge-cic/
File: edge-cic.kicad_sch

Title: FujiNet NES Rev0 — 72-pin cart edge, CIClone

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 2/8



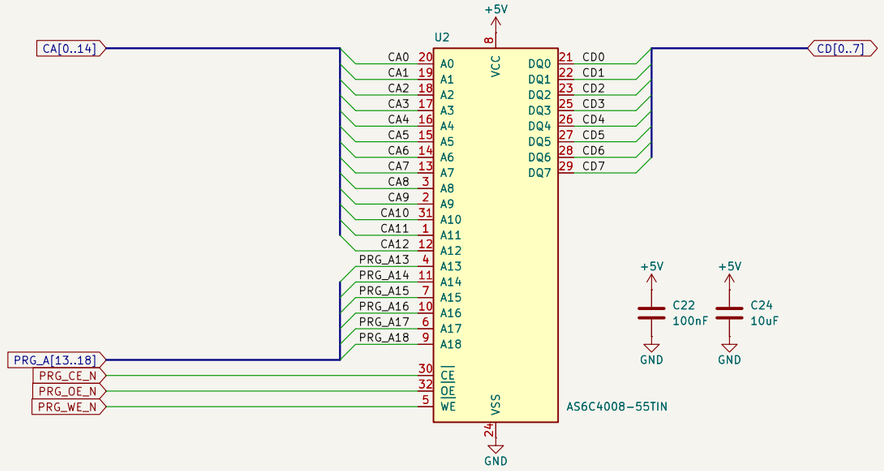
Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

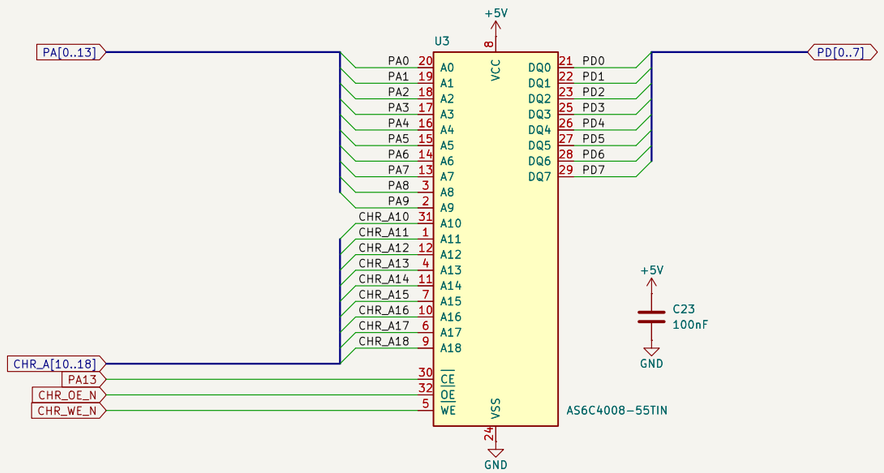
FujiNet

Sheet: /rp2354b/
File: rp2354b.kicad_sch

Title: FujiNet NES Rev0 — RP2354B bus controller

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 3/8



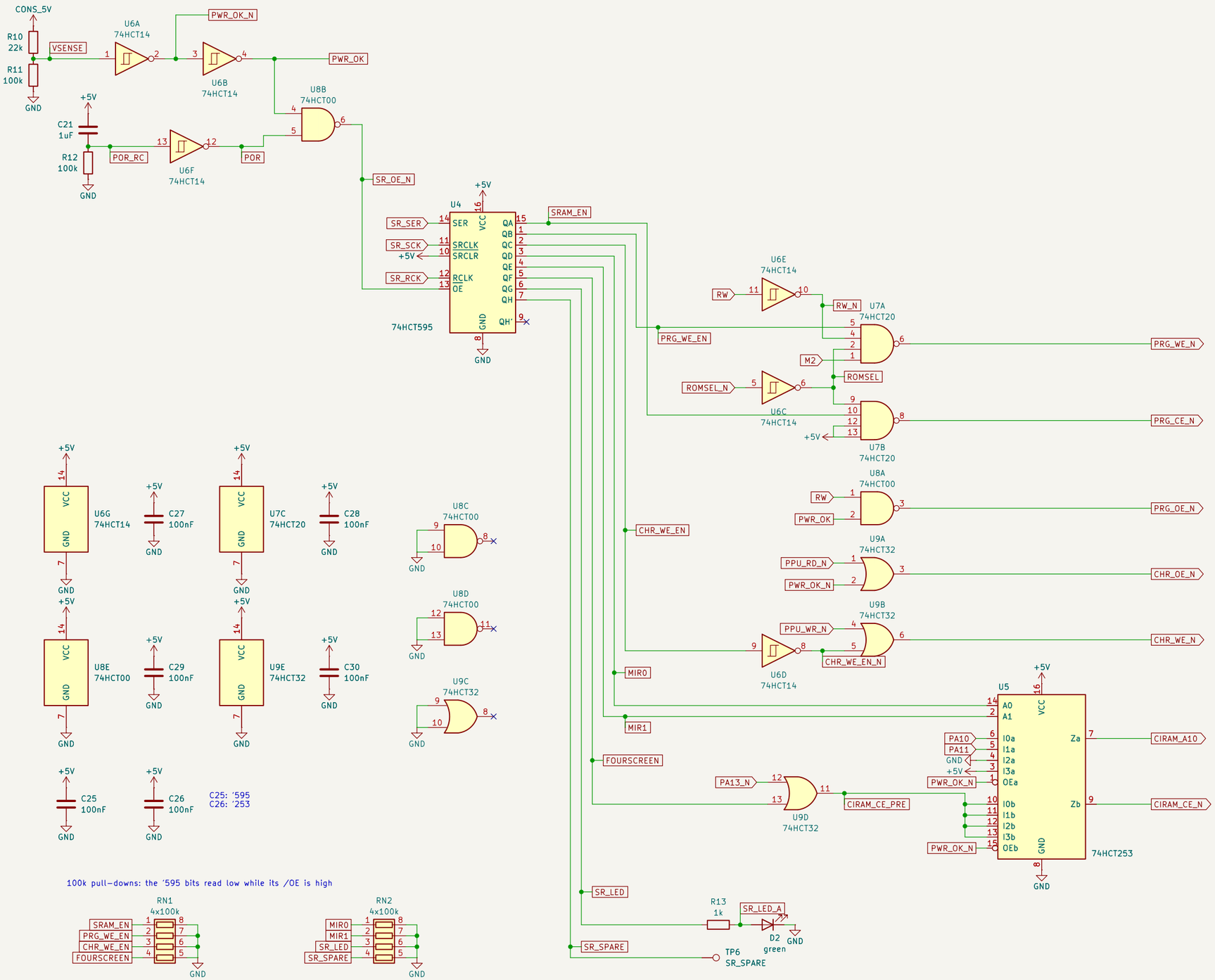


Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file
FujiNet

Sheet: /sram/
File: sram.kicad_sch

Title: FujiNet NES Rev0 — PRG and CHR SRAM

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 4/8



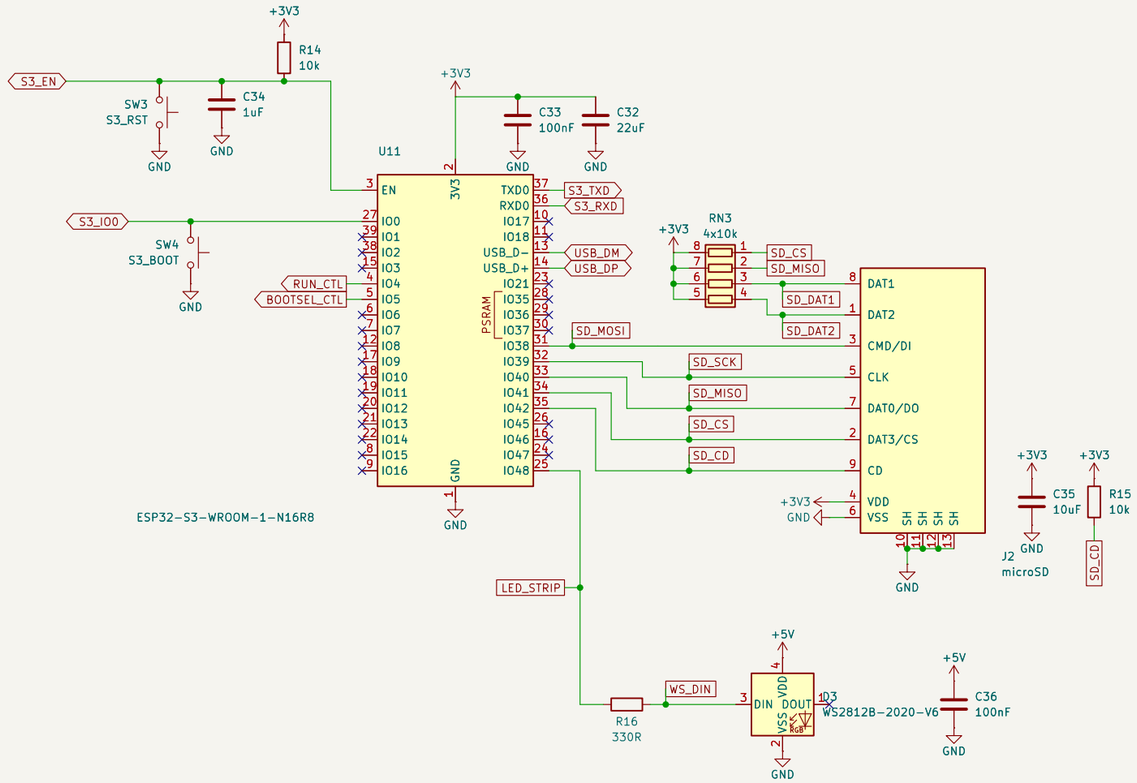
Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /glue/
File: glue.kicad_sch

Title: FujiNet NES Rev0 — 5V glue logic, ‘595 control bits

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 5/8



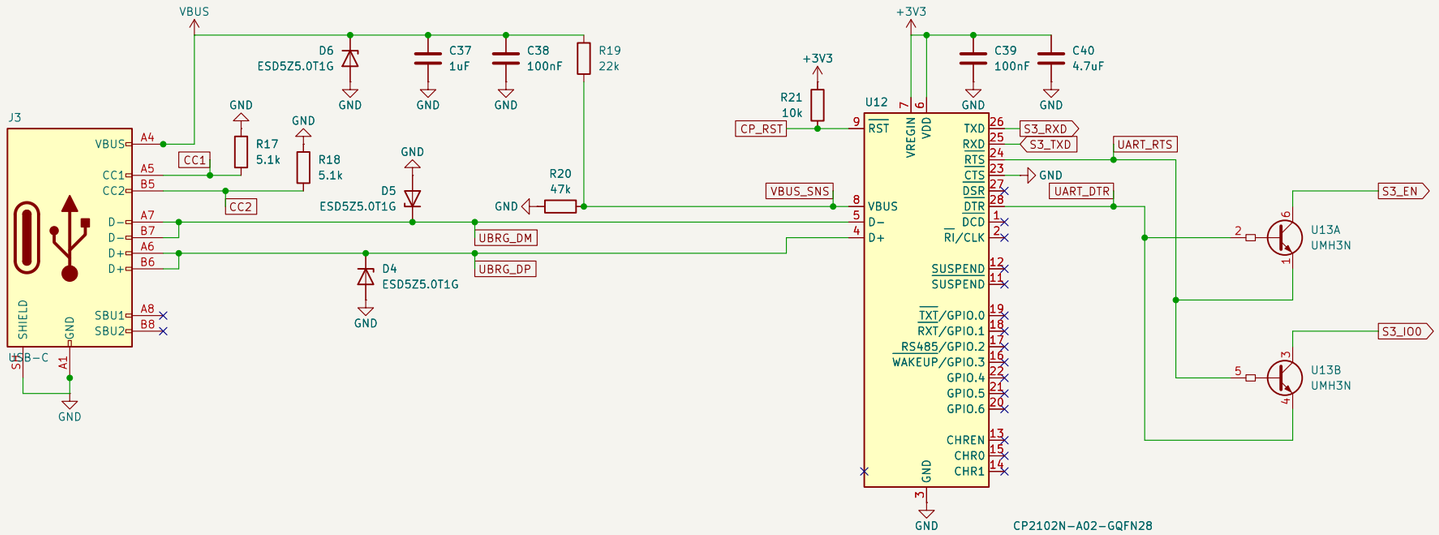
Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /esp32s3-sd/
File: esp32s3-sd.kicad_sch

Title: FujiNet NES Rev0 — ESP32-S3, microSD, status LED

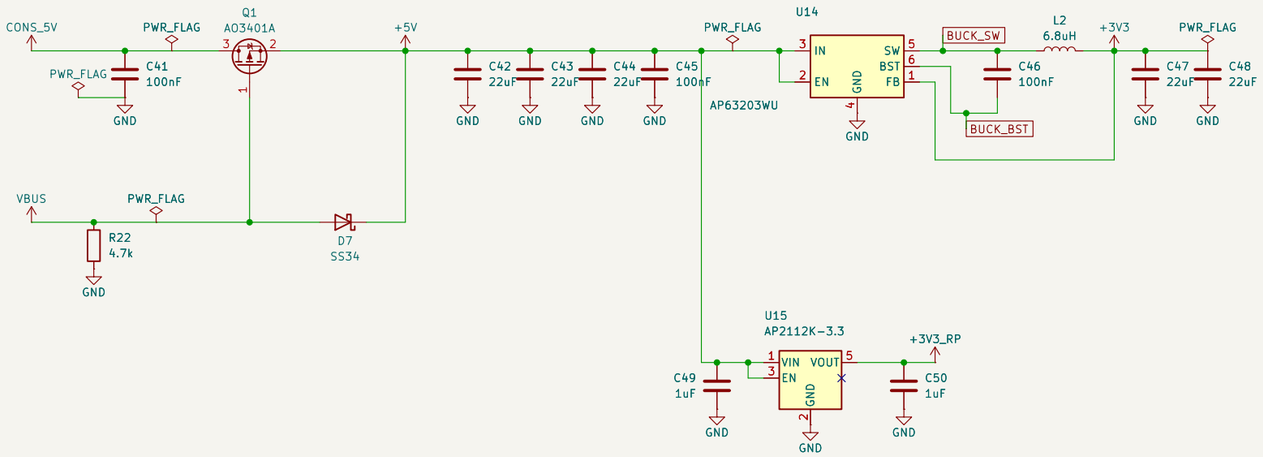
Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 6/8



Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /usb-uart/		
File: usb-uart.kicad_sch		
Title: FujiNet NES Rev0 — USB-C, CP2102N bridge		
Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 7/8



Rev0 audit 2026-10-01: kicad-happy + datasheets; see docs/design-review-rev0.md
CERN-OHL-W-2.0 (derived from FujiNet-Astrocade-Rev0 / FujiNet-INTV-Rev0 / PiNTY CARD)
Generated by tools/gen_sch.py + tools/sch_layout.py from tools/design.py — edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet NES Rev0 — Power: 5V OR, 3.3V buck, RP LDO

Size: A3	Date: 2026-10-04	**Rev: 0**
KiCad E.D.A. 10.0.6		Id: 8/8